Credit Risk and Default Prediction Model

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Introduction

This project develops a machine learning model to classify borrowers into credit risk categories and predict loan default using Lending Club loan data. The goal is to support more accurate, data-driven credit risk and default prediction model to assess creditworthiness by incorporating borrowers' key loan features and demographic data.

Data Preprocessing

The workflow for this project follows a typical end-to-end data science pipeline:

1. Dataset acquisition from Kaggle
2. Exploratory data analysis (EDA) and cleaning
3. Data Transformation and encoding and scaling
4. Data extraction into Python for analysis
5. Exploratory Data Analysis and Feature Investigation
6. Machine Learning model development for credit risk classification.
7. Deployment of the trained model within an interactive Streamlit dashboard

Key Findings

Exploratory data analysis and model results highlight several key factors associated with elevated borrower risk within the dataset:

- Payment history is a strong indicator of borrower risk.
- Younger borrowers appear more frequently in higher risk segments.
- Lower income levels are associated with increased default probability.
- Loan purposes such as consolidation and concentrate among higher risk borrowers.

These insights support the development of predictive models capable of identifying borrowers who pose a greater higher credit risk.

Dataset Overview

The dataset contains borrower financial and credit behavior variables commonly used in credit risk modeling.

These variables capture borrower income, credit exposure, debt levels, and repayment behavior, which are key indicators

The dataset contains information on 150,000 borrowers across multiple credit risk segments.

```
<class_name_ref_dataset_name_ref_dataset_name_ref_monthly_income_debt_ratio_ref_>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 12 columns):
#  Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
0   SeriousDlqin2yrs                      150000 non-null int64
1   RevolvingUtilizationOfUnsecuredLines  150000 non-null float64
2   age                                   150000 non-null int64
3   NumberOfTime30-59DaysPastDueNotWorse  150000 non-null int64
4   DebtRatio                             150000 non-null float64
5   MonthlyIncome                         120269 non-null float64
6   NumberOfOpenCreditLinesAndLoans       150000 non-null int64
7   NumberOfTimes90DaysLate               150000 non-null int64
8   NumberRealEstateLoansOrLines          150000 non-null int64
9   NumberOfTime60-89DaysPastDueNotWorse  150000 non-null int64
10  NumberOfDependents                    146076 non-null float64
dtypes: float64(6), int64(6)
memory usage: 13.7+ MB
```

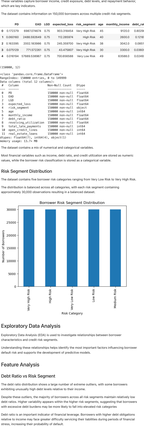
The dataset contains a mix of numerical and categorical variables.

Most financial variables such as income, debt ratios, and credit utilization are skewed as numerical features, while the borrower risk classification is stored as a categorical variable.

Risk Segment Distribution

The dataset contains five borrower risk categories ranging from Very Low Risk to Very High Risk.

The distribution is balanced across all categories, with each risk segment containing approximately 30,000 borrower records, ensuring sufficient representation for each risk category.

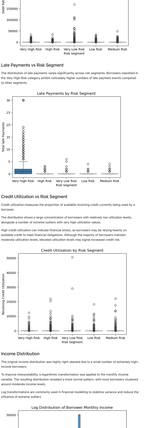


Debt Ratio by Risk Segment

Debt ratio increases significantly across higher risk categories, with the highest median and variance observed among Very High Risk borrowers.

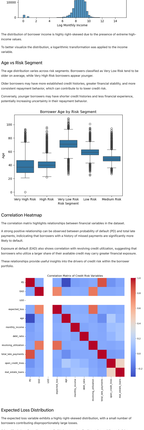
Despite these outliers, the majority of borrowers across all risk segments maintain relatively low debt ratios, higher variability appears within the higher risk segments, suggesting that borrowers with extensive debt facilities may be more likely to fall into elevated risk categories.

Debt ratio is an important indicator of financial leverage, borrowers with higher debt obligations relative to income may face greater difficulty servicing their liabilities during periods of financial stress, increasing their probability of default.



Late Payments vs Risk Segment

The distribution of late payments varies significantly across risk segments. Borrowers classified in the Very High Risk category exhibit noticeably higher numbers of late payment events compared to other segments.

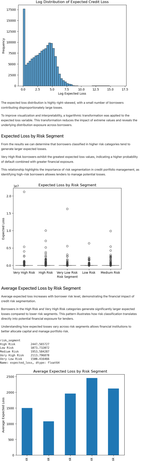


Credit Utilization vs Risk Segment

Credit utilization measures the proportion of available revolving credit currently being used by a borrower.

The distribution shows a large concentration of borrowers with relatively low utilization levels, particularly in number of borrowers in the Very Low Risk and Low Risk segments.

High credit utilization can indicate financial stress, as borrowers may be relying heavily on available credit to meet financial obligations. Although the majority of borrowers maintain moderate utilization levels, elevated utilization levels may signal increased credit risk.



Income Distribution

The original income distribution is highly right-skewed due to a small number of extremely high-income borrowers.

To improve interpretability, a logarithmic transformation was applied to the monthly income variable. The resulting distribution remained a more normal pattern, with most borrowers clustered around moderate income levels.

Using transformations are commonly used in financial modelling to stabilize variance and reduce the influence of extreme outliers

The distribution of borrower income is highly right-skewed due to the presence of extreme high-income values.

To better visualize the distribution, a logarithmic transformation was applied to the income variable.

Age vs Risk Segment

The age distribution varies across risk segments. Borrowers classified as Very Low Risk tend to be older on average, while Very High Risk borrowers appear younger.

Younger borrowers may have shorter credit histories, greater financial stability, and more consistent repayment behavior, which can contribute to lower credit risk.

Conversely, younger borrowers may have shorter credit histories and less financial experience, potentially increasing uncertainty in their repayment behavior.

Correlation Heatmap

The correlation matrix highlights relationships between financial variables in the dataset.

A strong positive relationship can be observed between probability of default (PD) and total loan exposure, indicating that borrowers with a history of missed payments are significantly more likely to default.

Exposure at Default (EAD) also shows strong correlation with revolving credit utilization, suggesting that borrowers utilize a larger share of their available credit may carry greater financial exposure.

These relationships provide useful insights into the drivers of credit risk within the borrower portfolio.

Expected Loss Distribution

The expected loss variable exhibits a highly right-skewed distribution, with a small number of borrowers contributing disproportionately large losses.

A logarithmic transformation was applied to improve visualization and reveal the underlying structure. The transformed distribution shows that most borrowers generate relatively small expected losses, while a minority contribute significantly higher loss exposure.

This loss pattern is typical in credit portfolios where a small number of high-risk borrowers account for a large share of potential losses.

The expected loss distribution is highly right-skewed, with a small number of borrowers contributing disproportionately large losses.

To improve visualization and interpretability, a logarithmic transformation was applied to the expected loss variable, revealing a more structured pattern across the majority of borrowers.

This transformed distribution ensures that most borrowers generate relatively small expected losses, while a minority contribute significantly higher loss exposure.

Expected Loss by Risk Segment

Expected loss increases across higher risk categories, with the highest median and variance observed among Very High Risk borrowers.

This aligns with the expectation that borrowers classified in higher risk categories tend to generate greater financial losses.

Higher expected loss in these portfolio segments reflects both higher default probability and higher exposure at default, underscoring the importance of credit portfolio management, as identifying high-risk borrowers allows lenders to manage potential losses.

Average Expected Loss by Risk Segment

Borrowers in the Very High Risk category have the highest average expected loss, demonstrating the financial impact of credit risk segmentation.

Exposure to the High Risk and Very High Risk categories generates significantly larger expected losses compared to Low Risk and Medium Risk. This pattern illustrates how risk classification translates directly into potential financial exposure for lenders.

Understanding how expected losses vary across risk segments allows financial institutions to better allocate capital and manage portfolio risk.

```
risk_segment
Very High Risk    2441.562137
High Risk         1111.788000
Low Risk           217.000000
Very Low Risk      111.798878
Medium Risk        -1.000000
Name: expected_loss, dtype: float64
```


Model Development

After exploring borrower characteristics and risk patterns in the dataset, the next step is to develop a borrower risk segmentation model capable of predicting borrower risk segments.

The objective of the model is to classify borrowers into credit risk categories using financial and behavioral variables.

This classification approach allows lenders to proactively manage credit exposure and identify high-risk borrowers before potential default occurs.

Model Selection

A Random Forest classifier was selected for this task due to its strong performance on structured tabular data.

```
              precision    recall  f1-score   support

           0       0.83      0.85      0.84      5015
           1       0.68      0.69      0.68      5000
           2       0.54      0.53      0.53      5000
           3       0.67      0.66      0.66      5000
           4       0.94      0.94      0.94      5000

    accuracy                           0.73     25015
   macro avg       0.73      0.73      0.73     25015
weighted avg       0.73      0.73      0.73     25015
```

Random Forest achieves a weighted average accuracy of approximately 88% in predicting borrower risk segments.

Precision, recall, and F1-scores remain consistently high across all risk categories, indicating that the model performs reliably when classifying borrowers across different levels of credit risk.

This performance suggests that the selected borrower characteristics provide strong predictive signals for assessing credit risk.

The confusion matrix provides a detailed view of the model's classification performance.

Most predictions fall along the diagonal of the matrix, indicating that the model correctly identifies the majority of borrower risk categories.

Misclassifications occur primarily between neighbouring risk segments, which is expected because adjacent risk categories often share similar borrower characteristics.

Feature Importance

This analysis reveals which borrower characteristics contribute most strongly to credit risk classification.

Age emerges as the most influential predictor, followed by total late payments and borrower income. These results suggest that age may serve as a proxy for a distinguishing between different borrower risk profiles.

Late payment history is particularly important, reinforcing the significance of repayment behavior as a key indicator of credit risk.